

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Kanmani L
Register Number	192511414

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

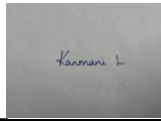
Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Kanmani L/192511414 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

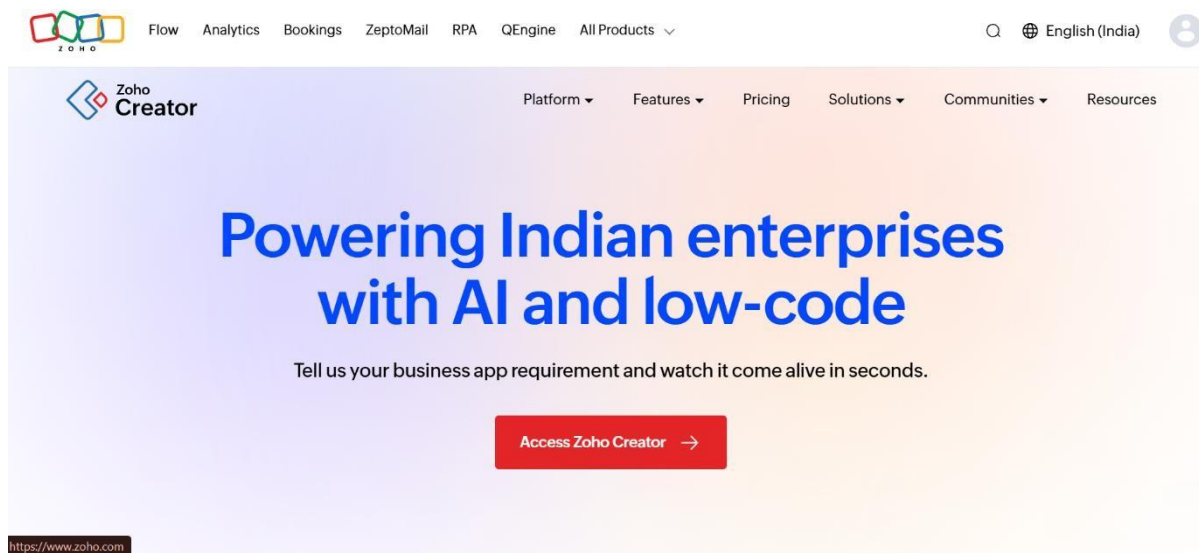
Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM:

To create a simple cloud software application for Library Book Reservation System for SIMATS Library using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

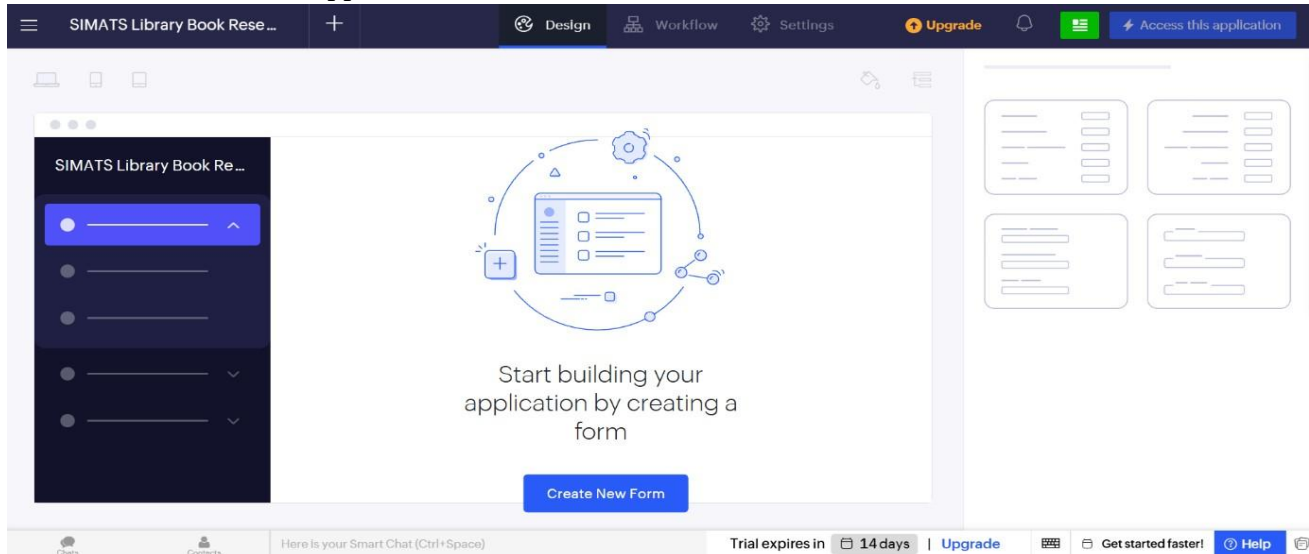
STEP 1: Log into zoho.com.



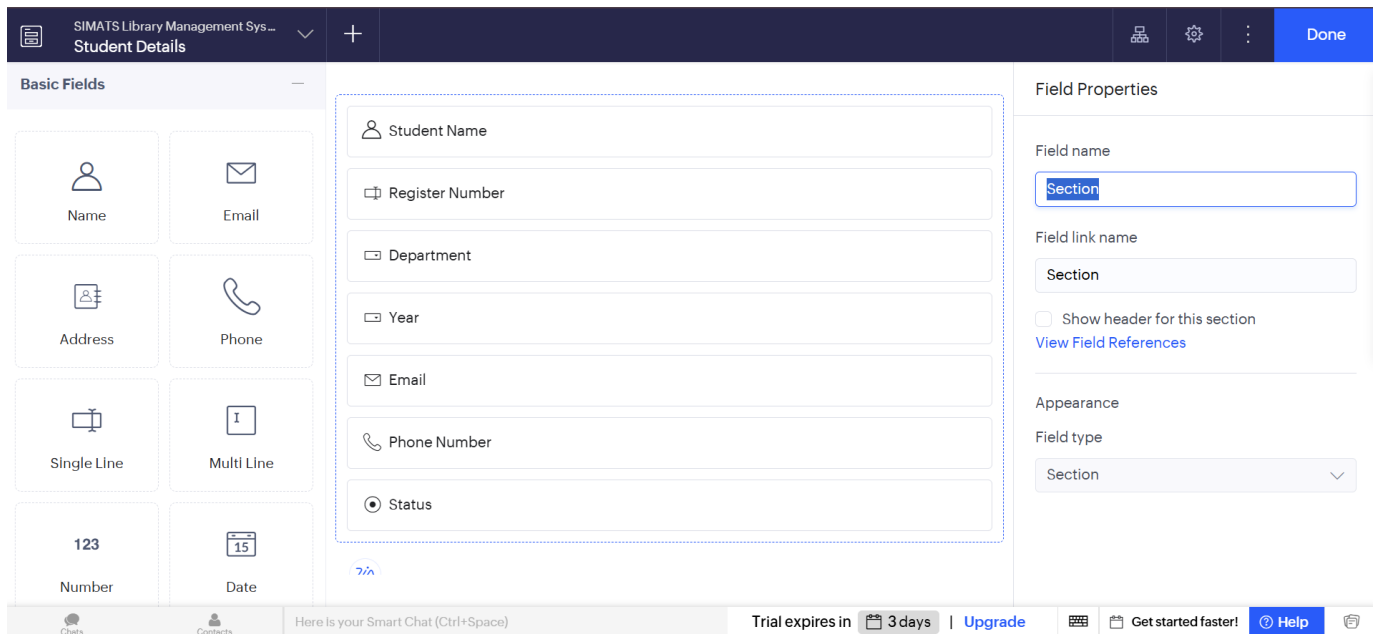
STEP 2: Create a new application from scratch in Zoho Creator.

The image shows the "Create from scratch" form in Zoho Creator. At the top, there are four icons: a blue circle, a white square, a white circle, and a white plus sign. Below these icons is the text "Create from scratch". The form has a light gray background. It contains a label "Application Name" with a red asterisk. Below the label is a text input field containing "SIMATS Library Book Reservation". Below the input field is a checkbox labeled "Enable Environment" with a link "Learn More" in blue. At the bottom of the form is a large blue button with the text "Create Application".

STEP 3: Create a new application and create forms from scratch.



STEP 4: Create Student Details Form



STEP 5: Create Book Form

The screenshot shows the 'Library Books' form creation interface. On the left, a 'Basic Fields' panel lists various field types: Name, Email, Address, Phone, Single Line, Multi Line, Number, and Date. The main workspace displays a list of fields for the 'Book' entity: Book ID, Title of the Book, Author, Publication, Year, Edition, Number of Copies, and Rack Number. The 'Book ID' field is selected, and its properties are shown on the right. The 'Field Properties' panel for 'Book ID' includes: Field name (Book ID), Field link name (book_id), View Field References link, Validation options (Mandatory, No duplicate values), Initial value (Initial value), and Character maximum (255). The bottom status bar indicates a trial expires in 3 days and provides links for Upgrade, Get started faster, and Help.

STEP 6: Create Book Reservation Form

The screenshot shows the 'Book Reservation' form creation interface. The 'Basic Fields' panel on the left is the same as in Step 5. The main workspace displays fields for the 'Book Reservation' entity: Student, Book, Reservation Date, Return Date, and Status. The 'Student' field is selected, and its properties are shown on the right. The 'Field Properties' panel for 'Student' includes: Field name (Student), Field link name (student), View Field References link, Display Fields (SIMATS Library Management System -> Student Details), and Display type (Drop Down). The bottom status bar indicates a trial expires in 3 days and provides links for Upgrade, Get started faster, and Help.

STEP 7: Create Book Issue Return Form.

The screenshot shows the 'Book Issue and Return' form creation interface. The 'Basic Fields' panel on the left is the same as in Step 5. The main workspace displays fields for the 'Book Issue and Return' entity: Student, Book, Issue Date, Return Date, and Status. The 'Student' field is selected, and its properties are shown on the right. The 'Field Properties' panel for 'Student' includes: Field name (Student), Field link name (student), View Field References link, Display Fields (SIMATS Library Management System -> Student Details), and Display type (Drop Down). The bottom status bar indicates a trial expires in 3 days and provides links for Upgrade, Get started faster, and Help.

STEP 8: Add sample data to all the created forms.

SIMATS Library Management System

Trial expires in 3 days Upgrade Edit this application Help

SIMATS Library Manage...

Dashboard

Book Reservations

+ Book Reservation

All Book Reservatio...

Statuses

Book Issue and Re...

Library Books

Student Details

Kanmani Log...

Book Reservation

Student

STU2026004

x

Book

1003

x

Reservation Date

08-Aug-2026

Return Date

10-Aug-2026

Status

Taken

x

Submit

Reset

SIMATS Library Management System

Trial expires in 3 days Upgrade Edit this application Help

SIMATS Library Manage...

Dashboard

Book Reservations

Book Issue and Re...

Library Books

+ Library Books

All Library Books

Statuses

Student Details

Kanmani Log...

Library Books

Book ID

1008

Title of the Book

Harry potter

Author

JK Rowling

First Name

Publication

dream publications

Year

2005

Edition

2

Number of Copies

5

Rack Number

4

Status

Available

x

Submit

Reset

STEP 9: Create reports to display the data entered in the respective forms.

SIMATS Library Management System

Trial expires in 3 days Upgrade Edit this application Help

SIMATS Library Manage...

Dashboard

Book Reservations

+ Book Reservation

All Book Reservatio...

Statuses

Book Issue and Re...

Library Books

Student Details

Kanmani Log...

All Book Reservations

Search + More

Student	Book	Reservation Date	Return Date	Status
STU2026004	1003	08-Aug-2026	10-Aug-2026	Taken
STU2026001	1004	10-Aug-2026	25-Aug-2026	Taken
STU2026005	1005	08-Aug-2026	22-Aug-2026	Returned
STU2026002	1002	05-Aug-2026	20-Aug-2026	Reserved
STU2026002	1003	01-Aug-2026	15-Aug-2026	Taken

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SIMATS Library Management System

Trial expires in 3 days Upgrade Edit this application Help

SIMATS Library Manage...

Dashboard

Book Reservations

Book Issue and Re...

Library Books

+ Library Books

All Library Books

Statuses

Student Details

Kanmani Log...

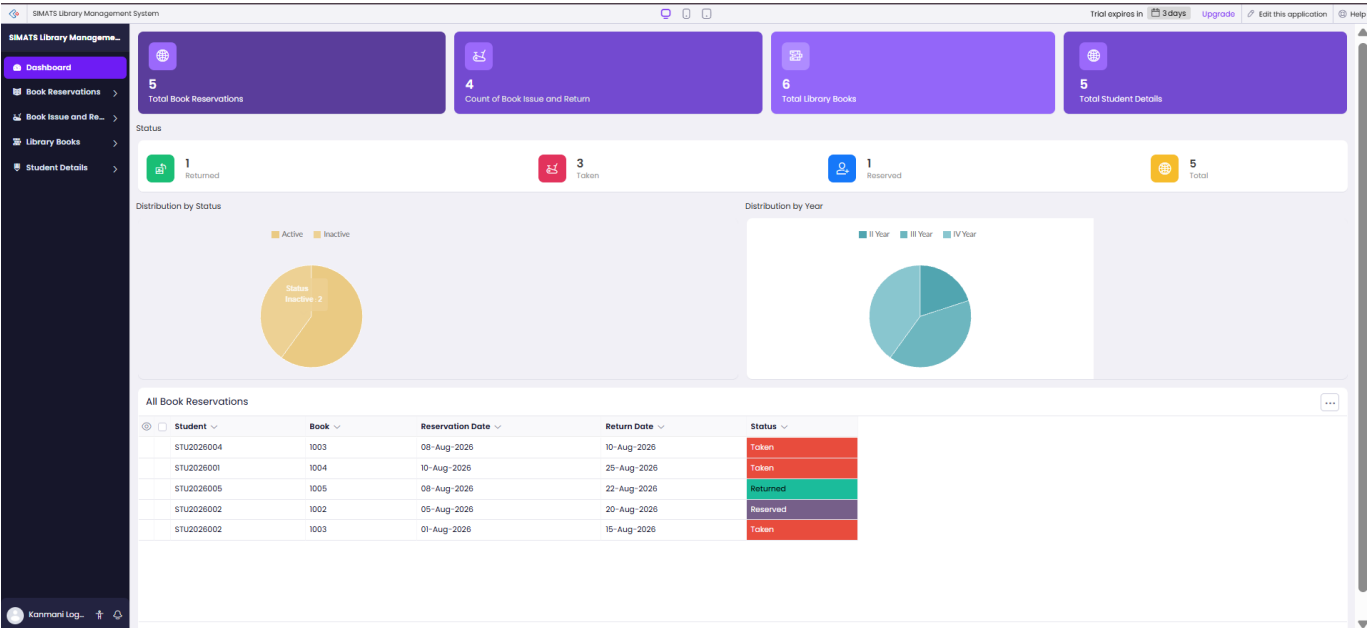
All Library Books

Search + More

Book ID	Title of the Book	Author	Publication	Year	Edition	Number of Copies	Rack Number	Status
1008	Harry potter	JK Rowling	dream publications	2005	2	5	4	Available
1005	Whispers in the Wind	J.K. Rowling	Bloomsbury	2025	1st Edition	6	E4	Reserved
1004	The Last Horizon	Ernest Hemingway	Modern Library	2021	3rd Edition	4	D1	Available
1003	Silent Echoes	Haruki Murakami	HarperCollins	2024	1st Edition	7	C3	Returned
1002	Mystery of the Lost Key	Agatha Christie	Vintage Books	2022	2nd Edition	3	B2	Available
1001	The Great Adventure	Jane Austen	Penguin Classics	2023	1st Edition	5	A1	Available

Showing 6 of 6

STEP 10: Create and display the Dashboard with important library information and book reservation details.



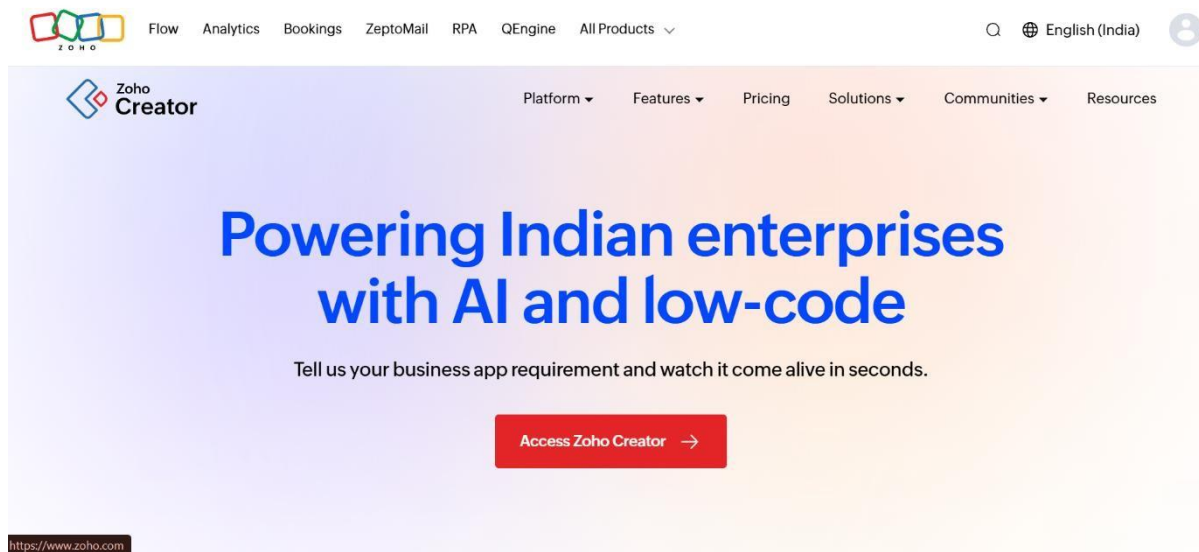
Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

To create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

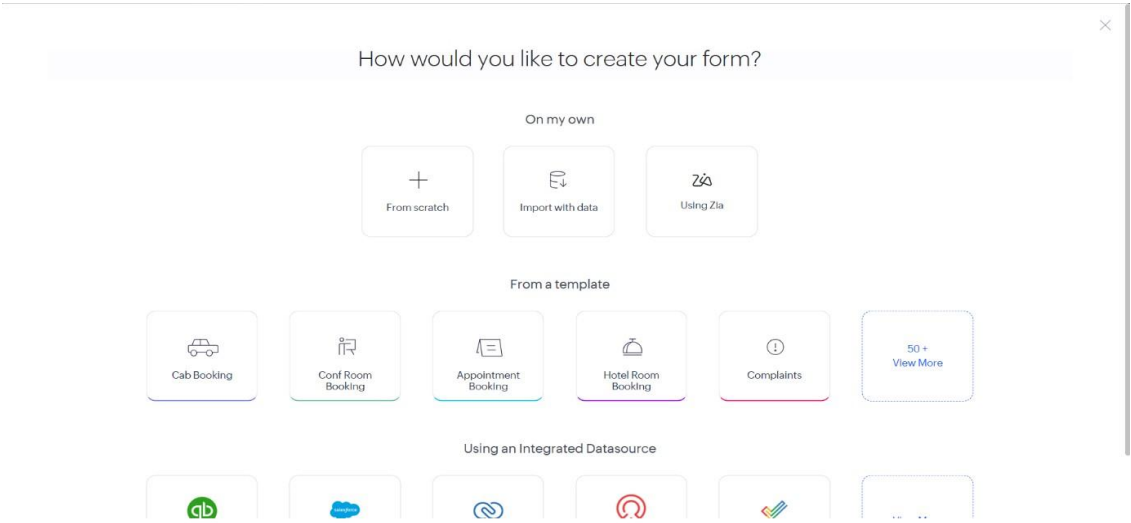
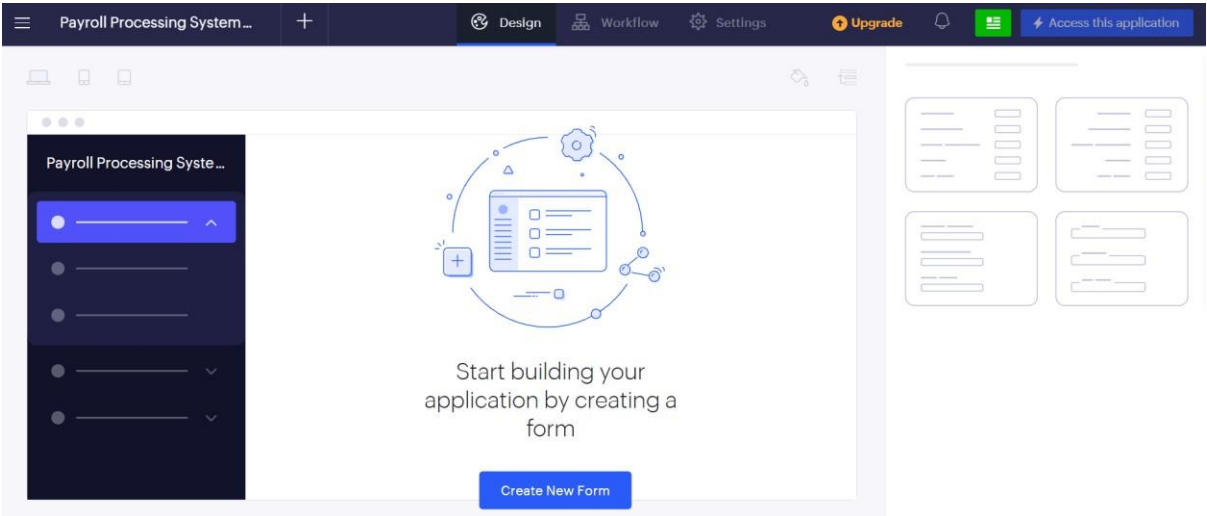
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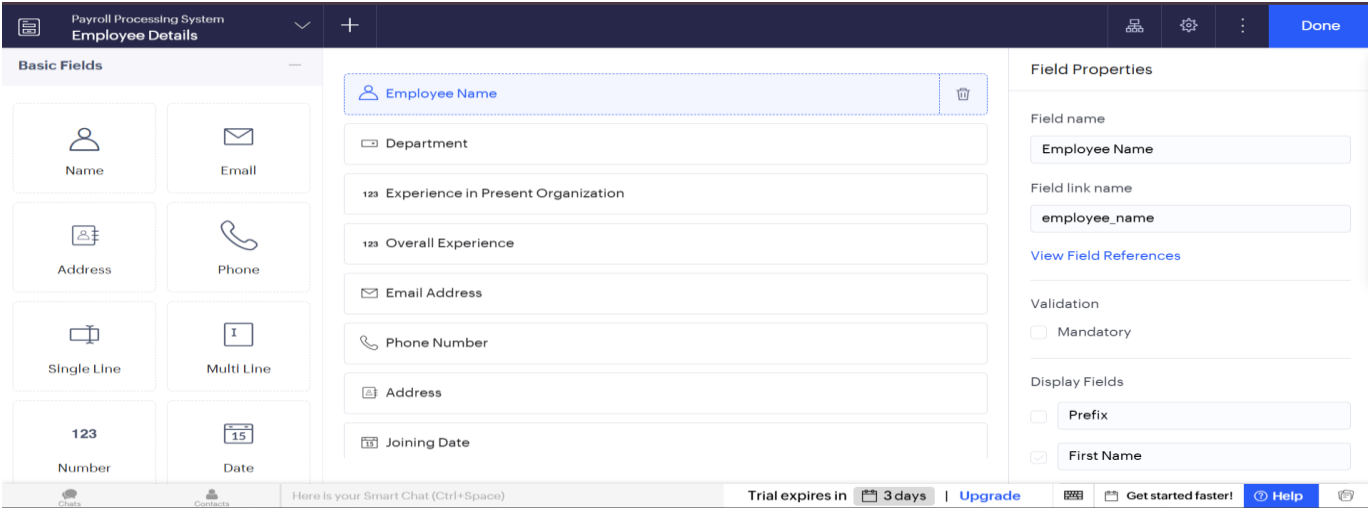
STEP 2: Create a new application from scratch in Zoho Creator.

A screenshot of the "Create from scratch" form in Zoho Creator. At the top, there are icons for creating from scratch (two circles and a plus sign). The form has a section titled "Application Name" with a red asterisk, containing a text input field with "Payroll Processing System". Below this is a checkbox labeled "Enable Environment" with a link "Learn More". At the bottom is a large blue button labeled "Create Application".

STEP 3: Create a new application and create forms from scratch.



STEP 4: Create Employee Details Form



STEP 5: Create Payroll Details Form.

Payroll Processing System

Payroll Details

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Employee

\$

Gross Pay

\$

Deductions

\$

Net Pay

Payroll Month

Deduction Type

Processing Date

Modified By

Field Properties

Field name

Notes

Field link name

notes

View Field References

Validation

Mandatory

Data Privacy

Contains personal data(PII)

User Input (for mobile)

Chats

Contacts

Here is your Smart Chat (Ctrl+Space)

Trial expires in 3 days | Upgrade

Get started faster!

Help

STEP 6: Create Payment Details Form

Payroll Processing System

Payment Details

Done

Basic Fields

Employee

\$

Net Pay Amount

Payment Status

Payment Date

Bank Name

Account Number

Transaction ID

Payment Mode

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Field Properties

Field name

Employee

Field link name

employee

View Field References

Display Fields

Payroll Processing System → Employee Details

First Name

+

-

+

Display type

Multi Select

Chats

Contacts

Here is your Smart Chat (Ctrl+Space)

Trial expires in 3 days | Upgrade

Get started faster!

Help

STEP 7: Create Salary Details Form

Payroll Processing System

Salary Details

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Date

Employee

Basic Pay

HRA

DA

TA

Total Allowances

Calculation Date

Field Properties

Field name

Employee

Field link name

employee

View Field References

Display Fields

Payroll Processing System → Employee Details

First Name

Display type

Multi Select

Trial expires in 3 days

Upgrade

Get started faster!

Help

STEP 8: Add sample data to all the created forms.

Payroll Processing System

Employee Details

Dashboard

Salary Details

Payment Details

Details

Employee Details

+ Employee Details

All Employee Details

Departments

Employee Name

Kanmani

First Name

Department

IT

Experience in Present Organization

3

Overall Experience

7

Email Address

kanmanilogu04@gmail.com

Phone Number

+91 8765789373

Address

123, t nagar, chennai

Address Line 1

Joining Date

08-Aug-2026

Employee Code

EMP0244

Status

Active

Inactive

Other

Kanmani Log...

STEP 9: Create reports to display the data entered in the respective forms.

Payroll Processing System

Dashboard

Salary Details

Payment Details

Details

Employee Details

Employee Details

All Employee Details

Departments

Kanmani Log...

Payroll Processing System

All Employee Details

Employee Name	Department	Experience in Present Organization	Overall Experience	Email Address	Phone Number	Address	Joini
Jennifer Lee	Marketing	5	7	j.lee@company.com	+18302844339	987 Ash Blvd	05-M
Michael Chen	Marketing	4	6	m.chen@company.com	+18607221988	654 Willow Way	10-Fe
Sarah Martinez	Finance	1	2	s.martinez@company.com	+17787644866	321 Spruce Dr	01-Au
Robert Williams	Finance	2	4	r.williams@company.com	+18308132800	456 Birch Ln	22-Ju
Alice Johnson	Marketing	3	5	alice.johnson@company.com	+17748667981	789 Cedar Ave	15-M
Kanmani	IT	3	7	kanmanilogu04@gmail.com	+918765789373	123, t nagar, chennai	08-A

Showing 6 of 6

STEP 10: Create and display the Dashboard

Payroll Processing System

Dashboard

Salary Details

Payment Details

Details

Employee Details

Kanmani Log...

\$ 350,120.00

Total Total Allowances

\$ 257,000.00

Total Basic Pay

\$ 64,250.00

Total HRA

\$ 23,070.00

Total DA

Currency Type

1

EUR

2

USD

2

INR

5

Total

Status by Overall Experience

Finance

IT

Marketing

Overall Experience

Active

Status

Inactive

Net Pay by Deductions

Deductions

Net Pay

All Salary Details

Employee	Basic Pay	HRA	DA	TA	Total Allowances	Calculation Date	Currency Type	Remarks	Create
Alice Johnson	\$ 42,000.00	\$ 10,500.00	\$ 3,780.00	\$ 2,520.00	\$ 59,800.00	08-Aug-2026	USD	New hire first month	Ac
Robert Williams	\$ 62,000.00	\$ 15,500.00	\$ 5,520.00	\$ 3,600.00	\$ 86,620.00	08-Aug-2026	INR	Senior staff allowance added	Hi
Alice Johnson	\$ 48,000.00	\$ 12,000.00	\$ 4,320.00	\$ 2,880.00	\$ 67,200.00	08-Aug-2026	USD	Regular monthly payment	Ac
Robert Williams	\$ 55,000.00	\$ 13,750.00	\$ 4,950.00	\$ 3,300.00	\$ 77,000.00	08-Aug-2026	INR	Annual bonus included	Hi
Sarah Martinez	\$ 50,000.00	\$ 12,500.00	\$ 4,500.00	\$ 3,000.00	\$ 69,500.00	08-Aug-2026	EUR	Monthly salary processed	Ac
Alice Johnson									
Sarah Martinez									
Michael Chen									
Jennifer Lee									

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Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

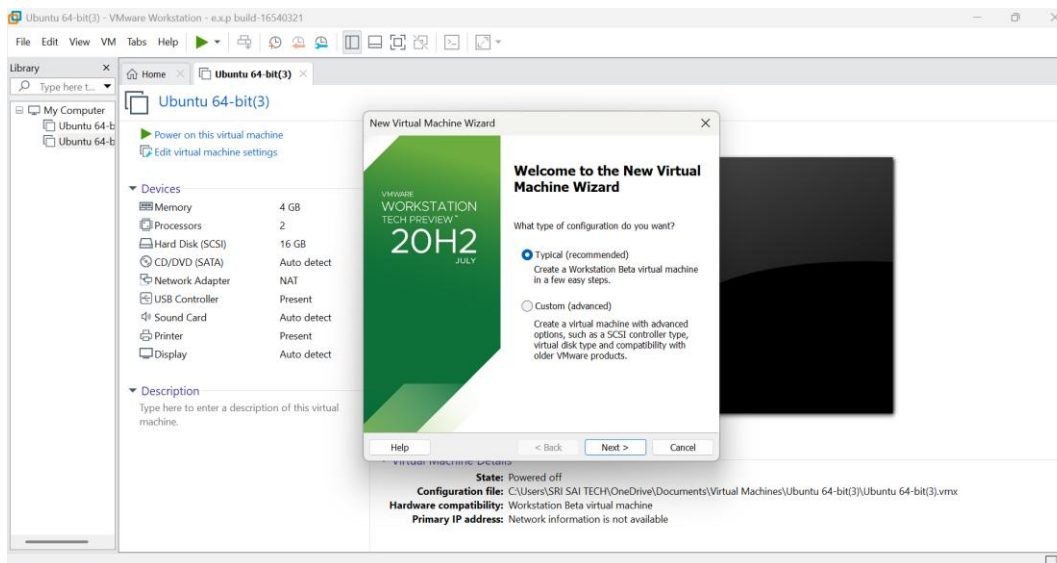
AIM:

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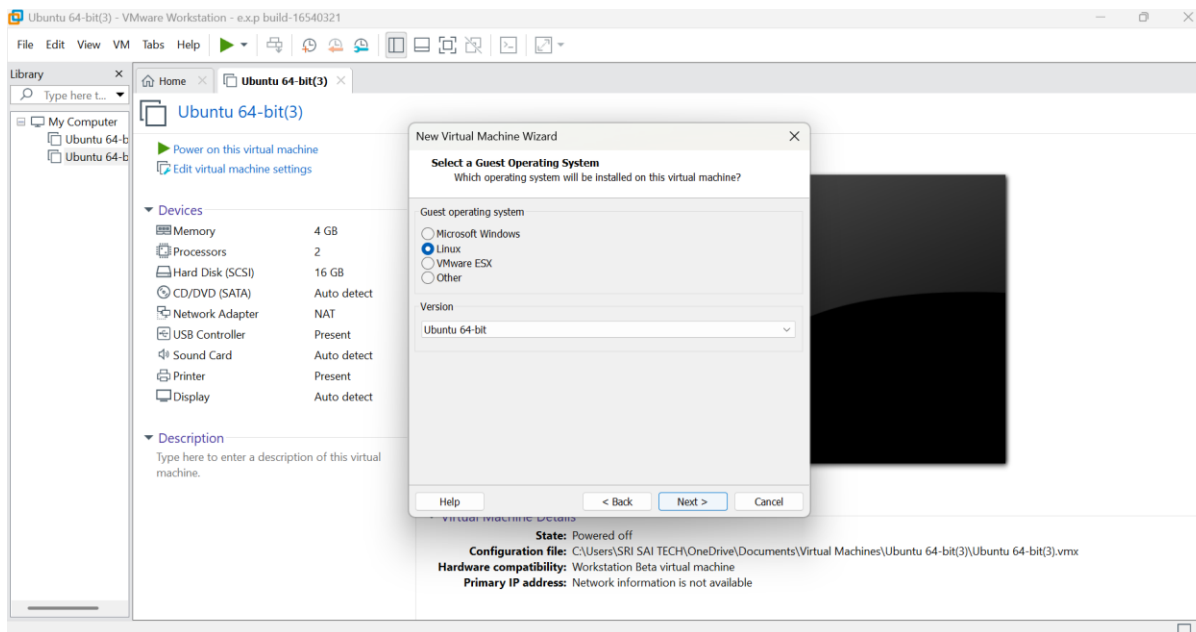
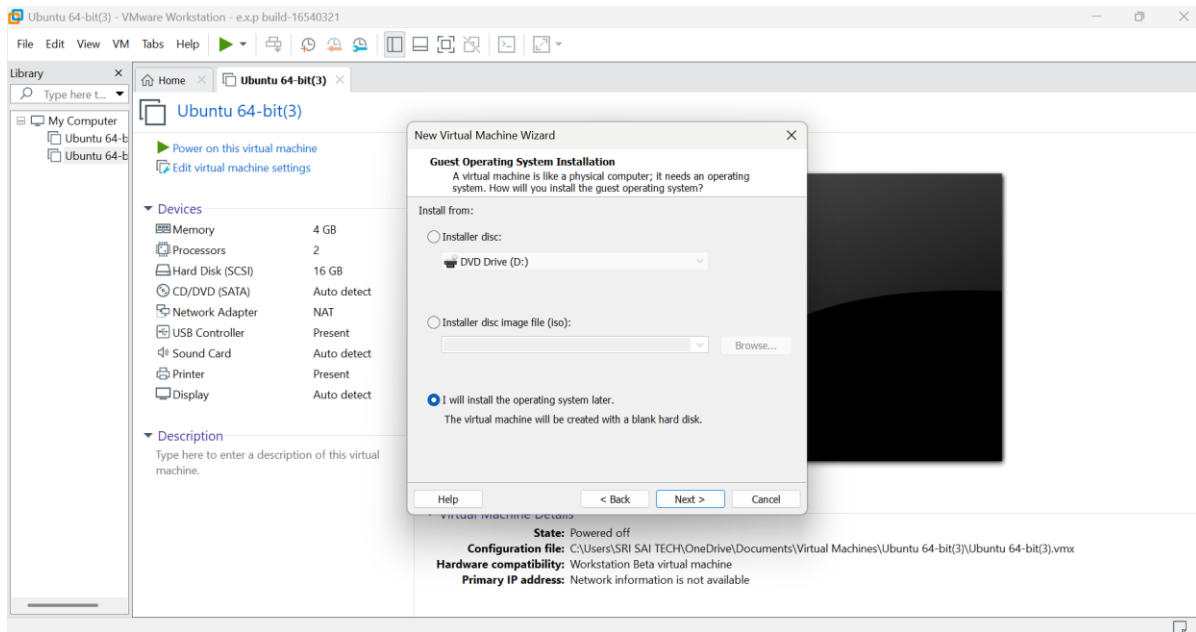
PROCEDURE:

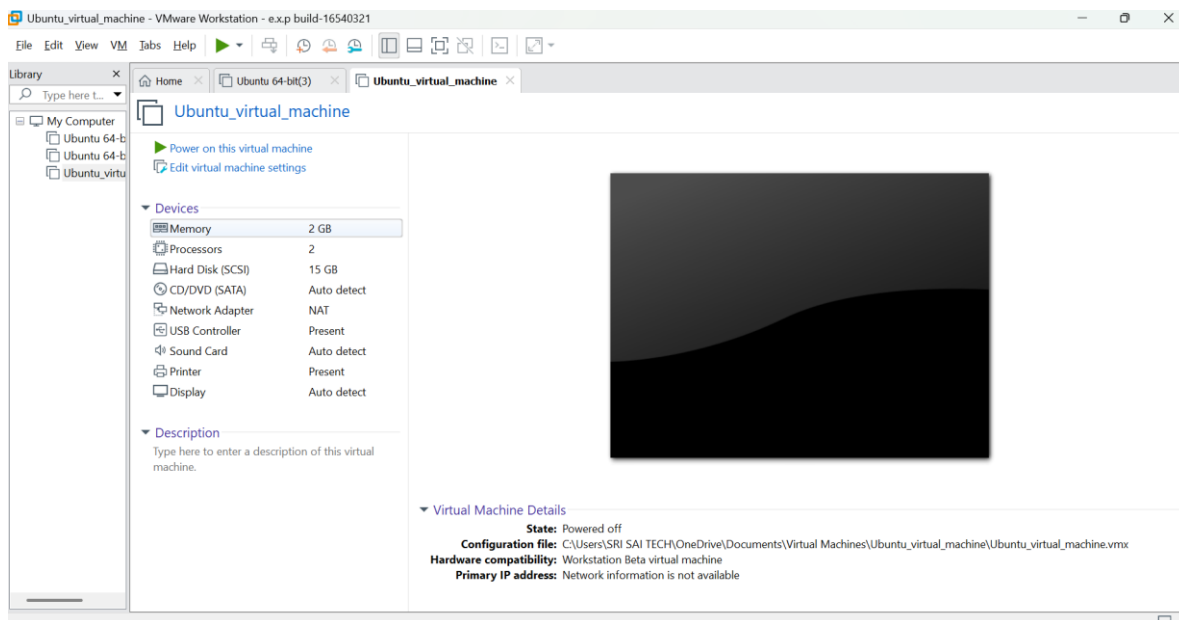
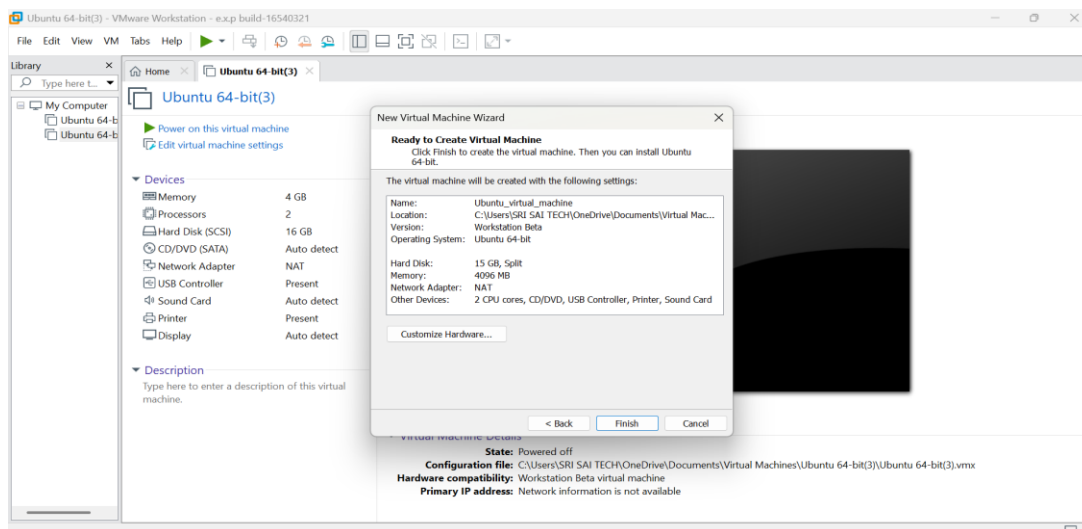
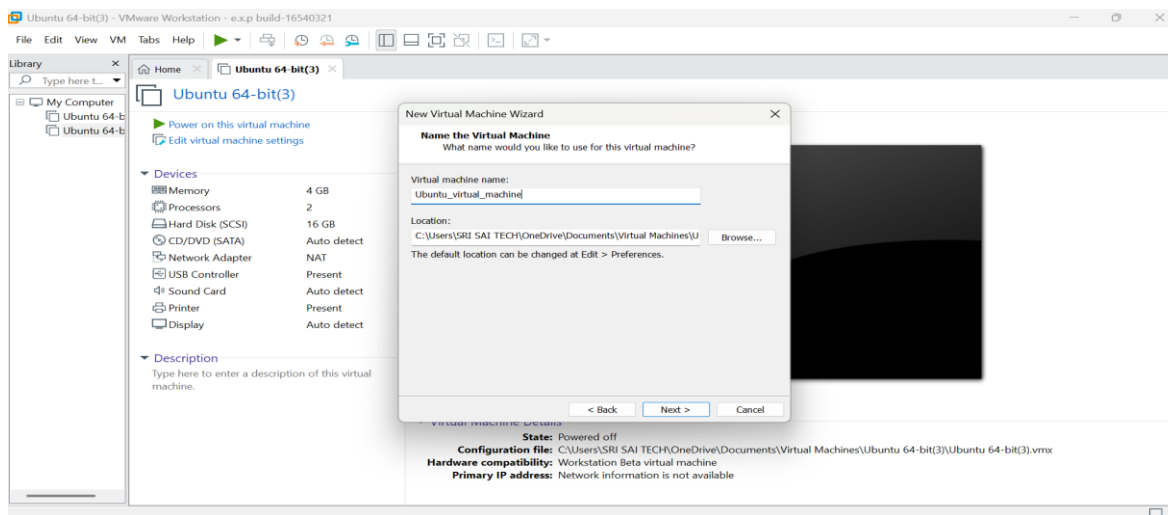
STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP 2: In VMware workstation->create new VM.

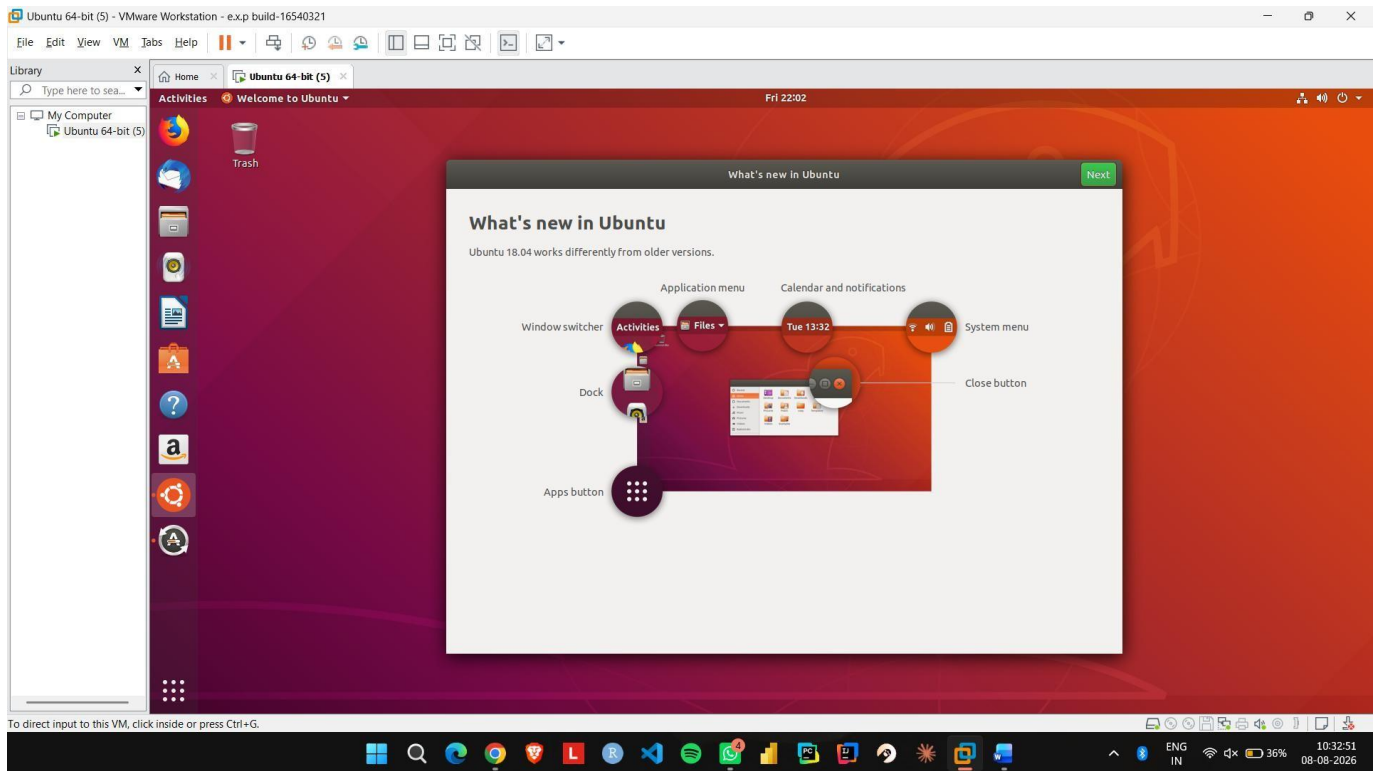


STEP 3: Do the basic configuration settings.





STEP 5: Create and Launch the VM.



Result

The virtual machine was successfully created using VMware with 1 CPU, 2 GB RAM and 15 GB storage.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

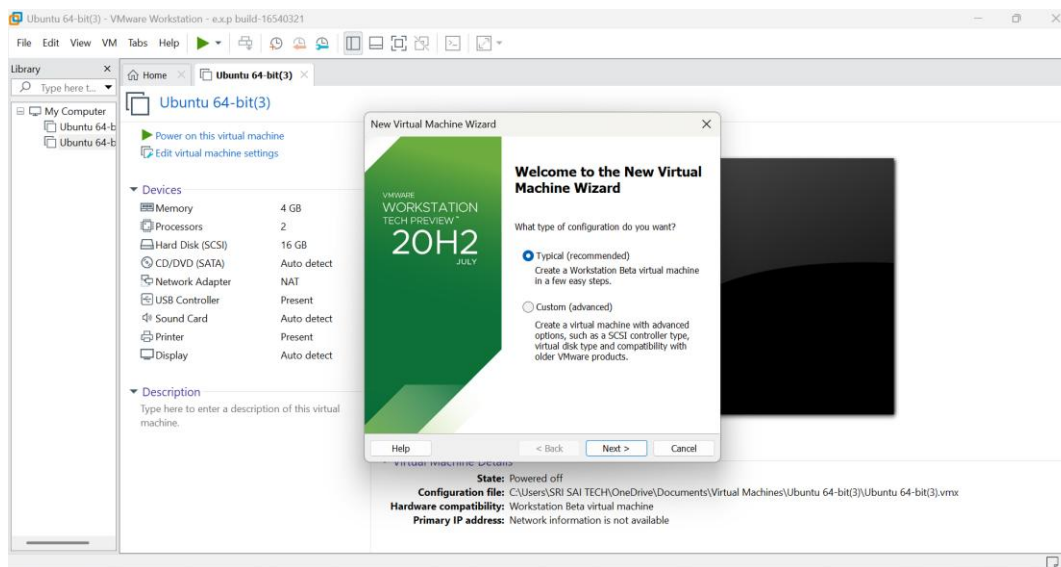
AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux) and to clone the VM and to test it

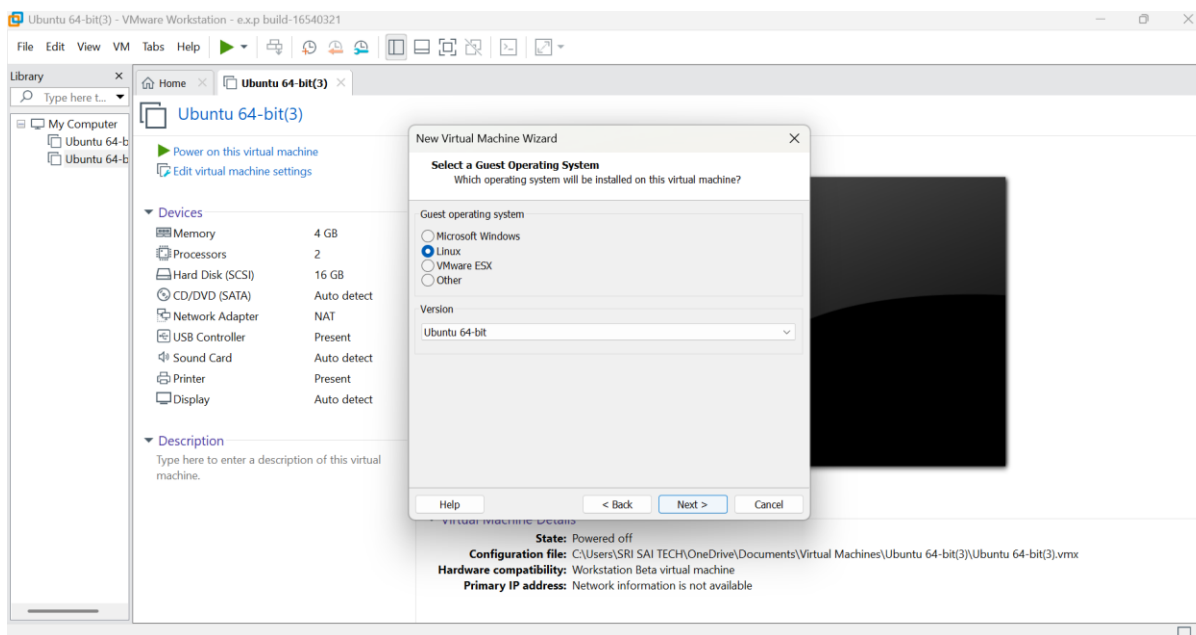
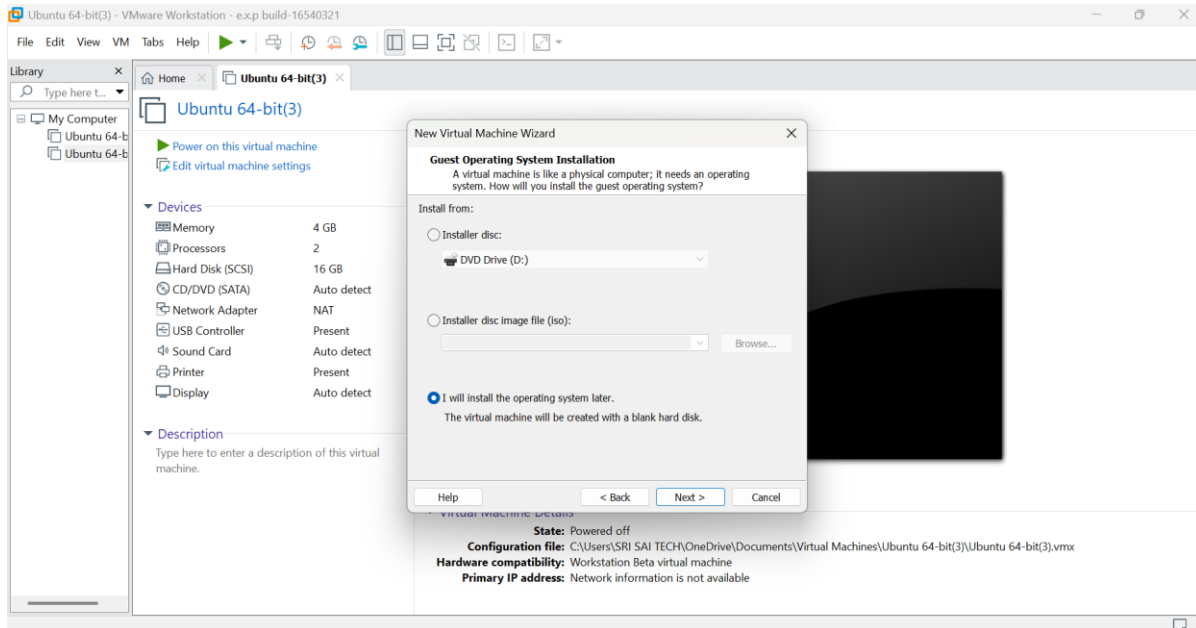
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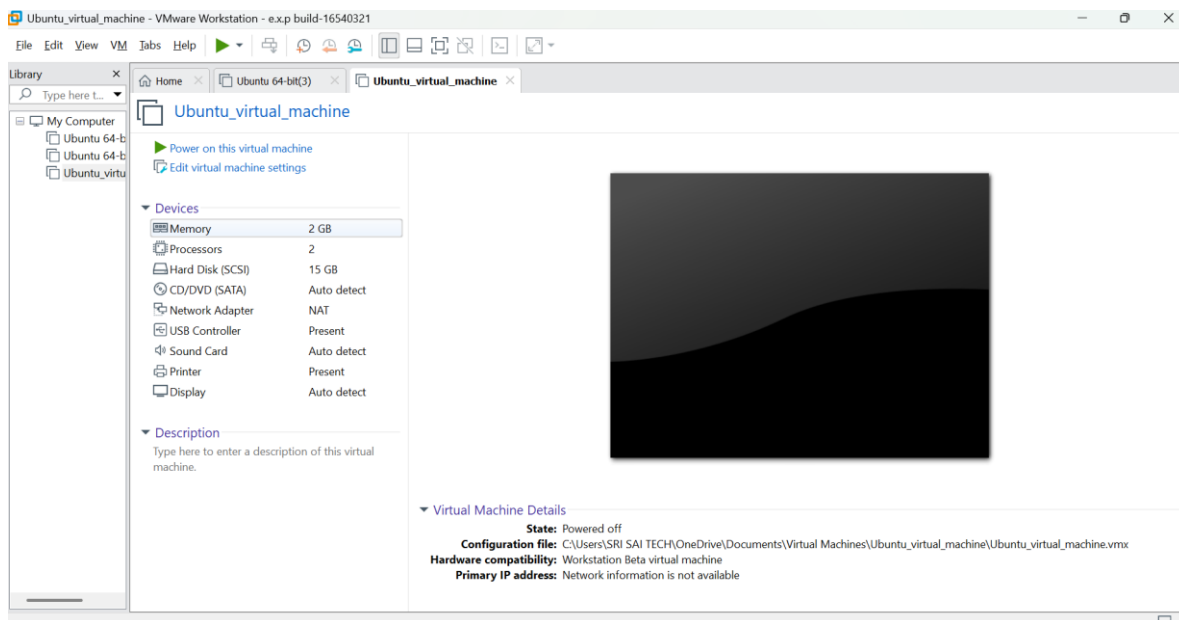
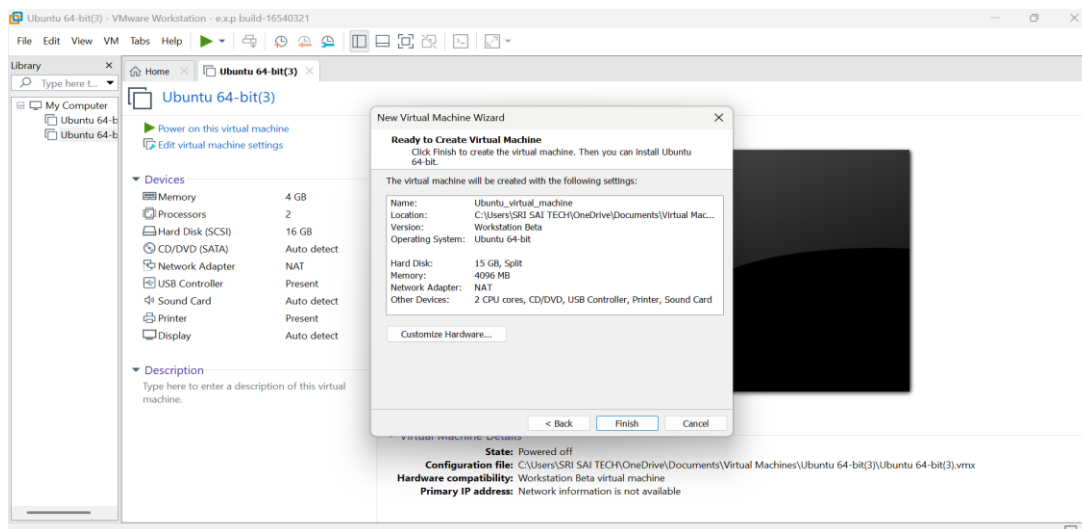
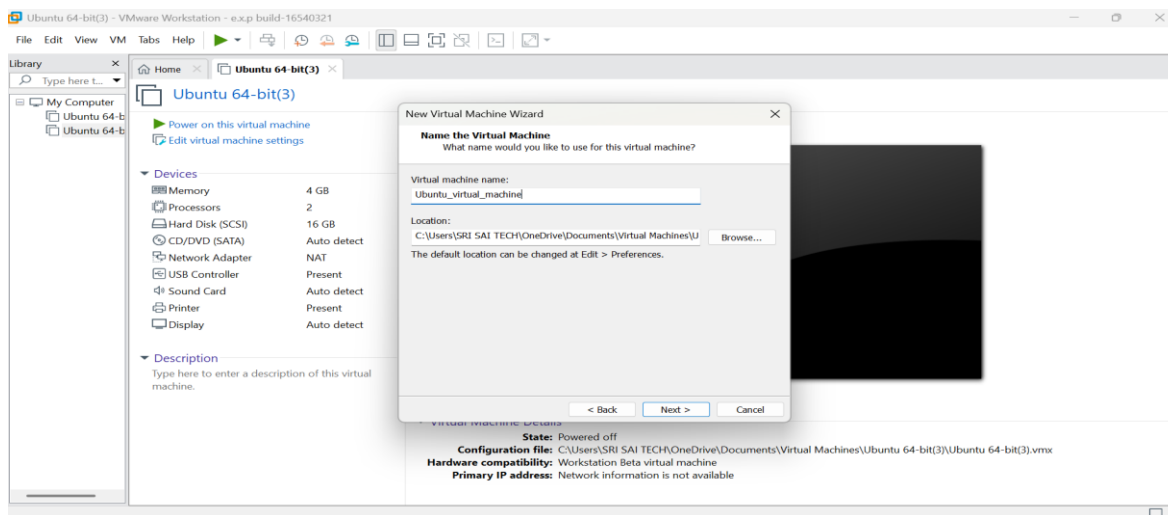
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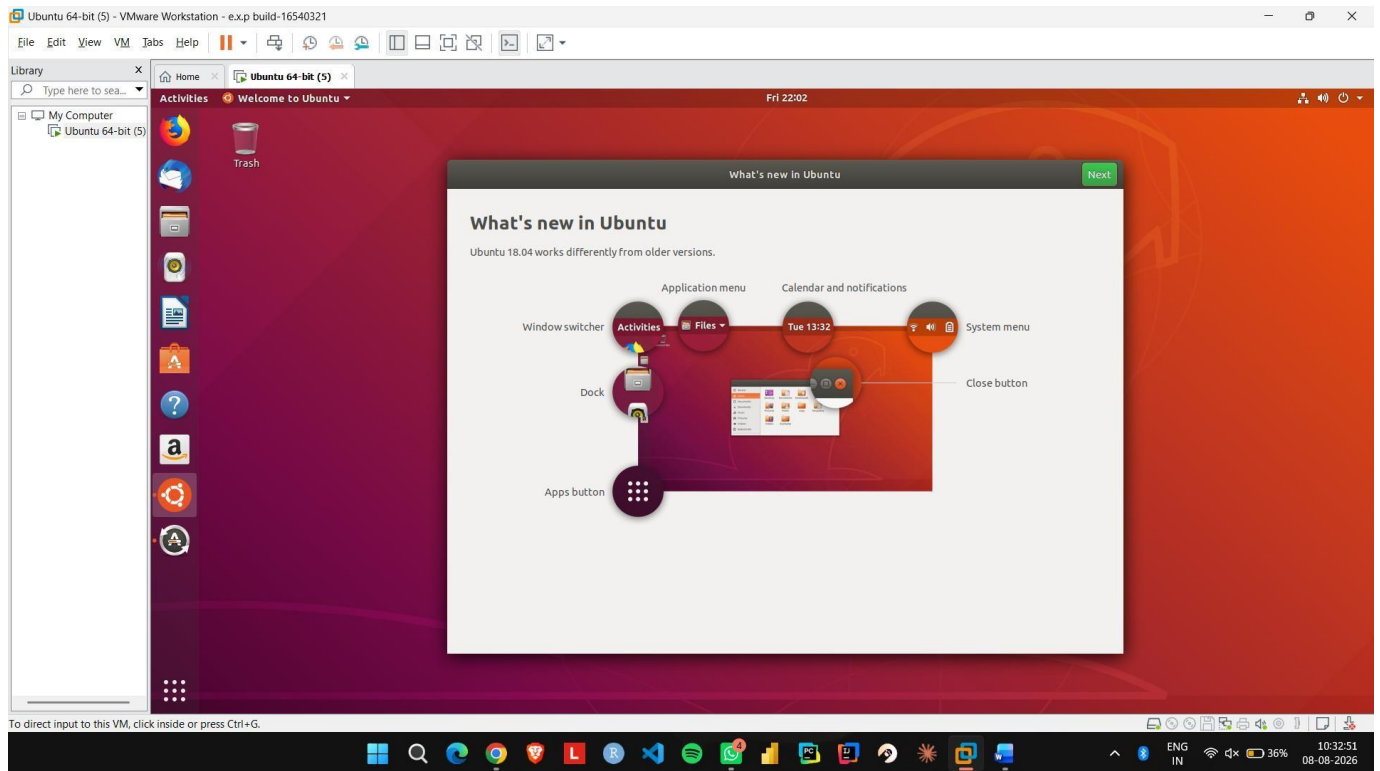


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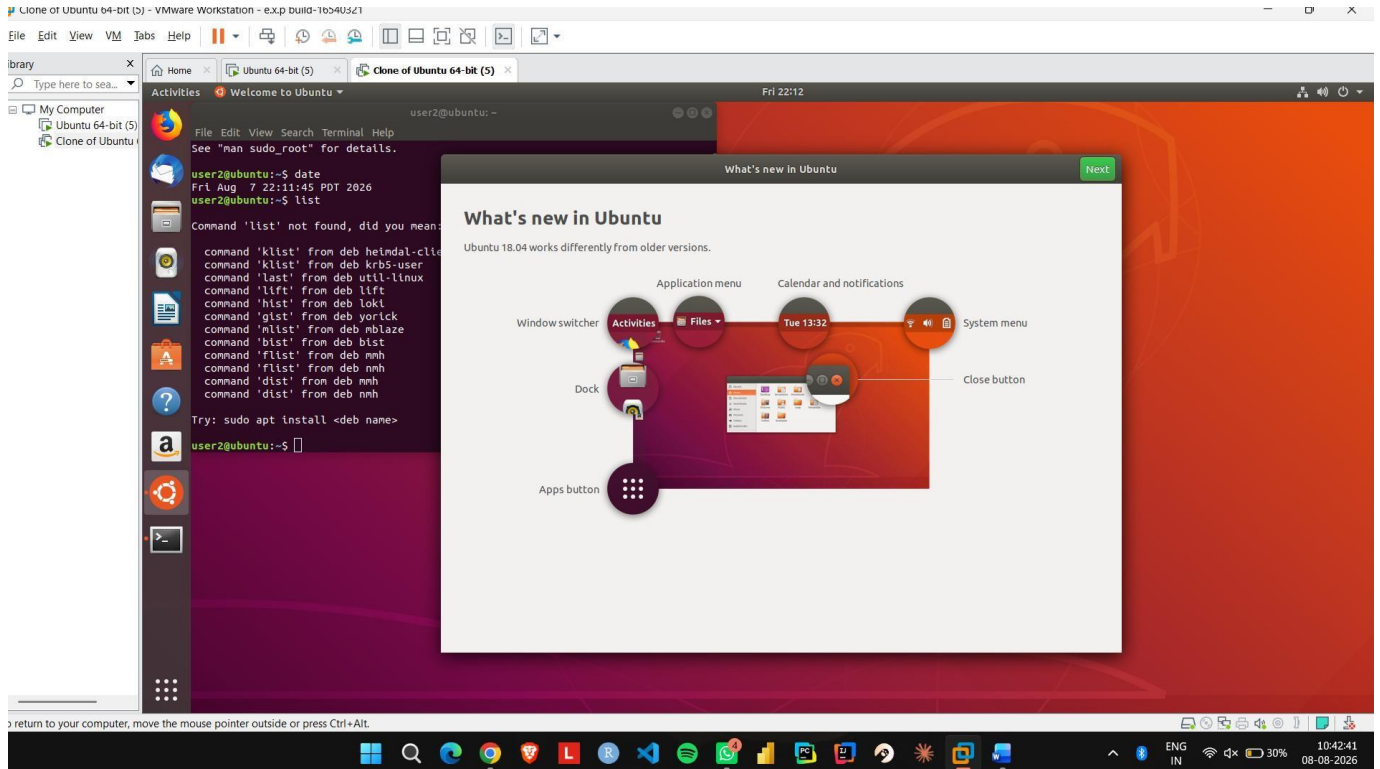




STEP 6: Create and Launch the VM.



STEP 7: Select Manage → Clone to start the cloning process



Result:

The VM was successfully cloned and tested, and its previous saved state was successfully restored